

	Option C follows the definition of amplitude of oscillation.
15	C Using $x = \lambda D/a$, $\lambda = (0.008)(0.0001) / (2.0) = 4.0 \times 10^{-7} \text{ m}$ Since the two sources are in phase at the slits and destructive interference occurs at the screen when their path difference = $(n + \frac{1}{2}) \lambda$ At 2nd order dark fringe, $n = 1$ Thus, path difference = $(1 + \frac{1}{2}) \lambda = 1.5 (4.0 \times 10^{-7}) = 6.0 \times 10^{-7} \text{ m}$
16	C